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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester B.Tech (CBCS) Degree Examinations, December 2025

25MAT103- CALCULUS AND DIFFERENTIAL EQUATIONS

(For CSE, ISE, CCE, AIM, AID, RAI, CYB, CBS)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

CO1: Q1 to Q4; CO2: Q5 to Q8; CO3: Q9 to Q12; CO4: Q13 to Q16; CO5: Q17 to Q20

- The polar form of the cartesian equation $x^2 + y^2 = 9$ is :
 A) $r = 3$ B) $r^2 = 18$
 C) $r^2 = 81$ D) $r = 9$
- If the curvature of a curve increases then, the radius of curvature _____
 A) increases B) constant
 C) decreases D) becomes zero
- The radius of curvature of the curve $y = e^x$ at the point $(0, 1)$ is:
 A) $\sqrt{2}$ B) $2\sqrt{2}$
 C) $\frac{\sqrt{2}}{2}$ D) 2
- The Maclaurin's series expansion of the function $f(x) = e^x$ is:
 A) $1 - x - \frac{x^2}{2} - \frac{x^3}{6} + \dots$ B) $1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$
 C) $1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \dots$ D) $-1 + x - \frac{x^2}{2} + \frac{x^3}{6} - \dots$
- If $z = 3x^2y + 5y$ then $\frac{\partial^2 z}{\partial x \partial y}$ is :
 A) $3xy$ B) $6xy$
 C) $3x + 5$ D) $6x$
- If $f(x, y) = \sin(xy + x^3y)$ then f_x at $(0, 1)$ is:
 A) 1 B) 0
 C) 2 D) 5
- If $f(x, y) = x^2y + x$ where $x = t$ and $y = t^2$ then the value of $\frac{df}{dt}$ when $t = 1$ is :
 A) 4 B) 1
 C) 2 D) 5
- Given $u = x + y, v = xy$ then the Jacobian $\frac{\partial(u, v)}{\partial(x, y)}$ is :
 A) 0 B) $x + y$
 C) $x - y$ D) 1
- The order and degree of the differential equation: $\frac{d^2y}{dx^2} = \frac{(\frac{dy}{dx})^2 + 1}{x}$ are _____ and _____.
 A) 2 and 3 B) 1 and 2
 C) 2 and 1 D) 2 and 2



10. The integrating factor (I.F.) for $\frac{dy}{dx} + \left(\frac{2}{x}\right)y = \sin x$ is:
A) e^{2x} B) x^2
C) $\frac{1}{x^2}$ D) x^3
11. The differential equation whose auxiliary equation has the roots 0, 1, -1 is:
A) $(D^3 - D^2 - D)y = 0$ B) $(D^3 - 2D^2 - D)y = 0$
C) $(D^3 + 2D^2 + D)y = 0$ D) $(D^3 - D)y = 0$
12. The solution of $y'' + y = 0$ is :
A) $y = C_1 \cos x + C_2 \sin x$ B) $y = C_1 e^x + C_2 e^{-x}$
C) $y = C_1 e^{ix} + C_2 e^{-ix}$ D) $y = C_1 x + C_2$
13. The partial differential equation of the expression $z = ax + by$, where a and b are arbitrary constants, is:
A) $z = pq$ B) $z = px + qy$
C) $z = p + q$ D) $z = px - qy$
14. The following second order partial differential equation is: $x^2 \frac{\partial^2 u}{\partial x^2} - x^2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$.
A) parabolic B) hyperbolic
C) elliptic D) circular
15. The order of PDE $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ is:
A) 1 B) 2
C) 0 D) 3
16. Which of the following is the standard form of the Laplace equation in 2D?
A) $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$ B) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$
C) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ D) $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y}$
17. The value of $\int_0^1 \int_0^x dy dx$ is
A) $\frac{1}{2}$ B) 1
C) 2 D) 0
18. Area of a region R in terms of polar co-ordinate is
A) $A = \iint_R dr d\theta$ B) $A = \iint_R r dr d\theta$
C) $A = \iint_R (r + 1) dr d\theta$ D) $A = \iint_R r^2 dr d\theta$
19. The value of $\Gamma\left(\frac{1}{2}\right) =$
A) $\sqrt{\frac{\pi}{2}}$ B) $\frac{\pi}{2}$
C) $\frac{\sqrt{\pi}}{2}$ D) $\sqrt{\pi}$
20. The value of $\beta(3, 2)$ is:
A) $\frac{1}{24}$ B) $\frac{1}{10}$
C) $\frac{1}{12}$ D) $\frac{1}{2}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Determine the angle of intersection between the polar curves:
$r = \frac{a\theta}{1+\theta}$ and $r = \frac{a}{1+\theta^2}$. | 6 | L3 | 1 | 2 |
| b) Obtain Taylor's series expansion of $\log x$ about $x = 1$ up to three non-vanishing terms. | 6 | L2 | 1 | 1 |

c)	With usual notations prove that $\frac{1}{p^2} = \frac{1}{r^2} \left[1 + \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 \right]$ where $p = r \sin \phi$.	4	L3	1	2
2. a)	Using standard polar coordinates, prove that $\tan \phi = r \frac{d\theta}{dr}$.	6	L3	1	2
b)	Find the radius of curvature at any point on the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$.	6	L3	1	2
c)	Expand the function $\cos x$ as a Maclaurin's series up to 3 nonvanishing terms.	4	L2	1	1

Unit – II

3. a)	If u is a homogeneous function of degree n in x and y , then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$.	6	L3	2	2
b)	Determine the volume of the greatest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.	6	L3	2	2
c)	Let $u(x, y) = x^2 e^y + y^2 \sin x$. Find the first and second order partial derivatives of u with respect to x and y .	4	L2	2	1
4. a)	Show that if $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, then $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.	6	L3	2	1
b)	If $x = r \cos \theta$, $y = r \sin \theta$, find the Jacobian $J = \frac{\partial(x, y)}{\partial(r, \theta)}$ and $J' = \frac{\partial(r, \theta)}{\partial(x, y)}$, and show that $JJ' = 1$.	6	L3	2	2
c)	Given $u = x^2 - y^2$, where $x = 2r - 3s + 4$, $y = -r + 8s - 5$, find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial s}$.	4	L2	2	2

Unit – III

5. a)	Determine the solution of the differential equation $\frac{d^2 y}{dx^2} + 4y = 2 \sin^2 x$.	6	L3	3	1
b)	Apply the method of variation of parameters to determine the solution of $(D^2 - 4D + 4)y = \frac{e^{2x}}{x}$.	6	L3	3	2
c)	Write the general solution of the differential equation: $(x^2 - 4xy - 2y^2) dx + (y^2 - 4xy - 2x^2) dy = 0$.	4	L2	3	1
6. a)	Determine the solution of the differential equation: $x \frac{dy}{dx} + y = -2x^6 y^4$.	6	L3	3	2
b)	Solve the differential equation $\frac{d^2 y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4$.	6	L3	3	1
c)	Find the general solution of the differential equation $(D^3 + 1)y = 0$.	4	L2	3	1

Unit – IV

7. a)	Form a PDE by eliminating the arbitrary function from $x + y + z = f(x^2 + y^2 + z^2)$.	6	L3	4	2
b)	Determine the analytical solution of Laplace equation of the form $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ by separation of variables.	6	L3	4	2

c)	Find the solution of the PDE by direct integration: $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y} + a$.	4	L2	4	1
8. a)	Solve the following higher order partial differential equations by direct integration: $\frac{\partial^2 z}{\partial x^2} + z = 0$, given that when $x = 0, z = e^y$ and $\frac{\partial z}{\partial x} = 1$	6	L3	4	1
b)	Apply the method of separation of variables to solve the PDE $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$.	6	L3	4	2
c)	Construct partial differential equations by eliminating the arbitrary constants from: i) $z = ax + by$ ii) $z = ax + by + a^2 + b^2$	4	L2	4	1
Unit – V					
9. a)	Determine the area of the region bounded between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$.	6	L3	5	2
b)	Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.	6	L3	5	2
c)	Find the value of the double integral: $\int_0^1 \int_0^{x^2} e^{\frac{y}{x}} dx dy$.	4	L2	5	1
10. a)	Change the order of integration and hence evaluate the following double integral: $\int_0^\infty \int_0^x x e^{-x^2/y} dy dx$.	6	L3	5	2
b)	Evaluate the following triple integral: $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) dy dx dz$	6	L3	5	1
c)	Find the value of $\int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta$ using Gamma function.	4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
